

MICROCANONICAL PREPARATION AND EXPONENTIAL-SCALE ENSEMBLE TRANSFER FOR DETERMINISTIC HARD SPHERES

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ABSTRACT. Microcanonical preparation is derived from the full connected hard-sphere source pressure, not an ideal compound proxy. Conditional regenerative full-rank blocks give a global mixed lattice-continuous characteristic theorem, typical activity gives uniform covariance, and an exact source-dependent saddle is retained. The shell likelihood remains inside the Gaussian coefficient, yielding the prepared pressure, LDP, and covariance Schur complement at every declared shell scale.

Platform identifier: HSBG-LI0-v3. **Controlling revision:** ROUND9-REFEREE-POSITIVE-CLOSURE.

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1. ROUND-NINE RECONSTRUCTION: THE FULL SOURCE PRESSURE AND A WEIGHTED EXACT-SADDLE SHELL COEFFICIENT

The compound-Poisson gas is used only to expose the nondegenerate one-particle mechanism. The finite hard-sphere pressure contains in addition the connected hard-core and trajectory remainder proved in the grand-canonical part of B2. All saddle, Fourier, and shell estimates below are established for that full pressure. Moreover the source-dependent likelihood is kept inside the shell integral; it is not replaced by its value at the shell centre when the shell has an extensive width.

Date: August 31, 2026.

2020 Mathematics Subject Classification. Primary 82C40, 60F10; Secondary 76P05, 82C22.

Key words and phrases. hard spheres, microcanonical ensemble, Liouville measure, information projection, Boltzmann-Grad limit.

1.1. The exact grand-canonical source pressure. Let H denote the bounded particle-path/contact source from B2 and let $\lambda \in \mathbb{R}^d$ couple to the constraint mark

$$X(z) = (1, v, |v|^2/2, \chi_1(z), \dots, \chi_m(z)).$$

For the hard-sphere law at scale ε , with $\mu_\varepsilon \rightarrow \infty$, put

$$Q_\varepsilon(H, \lambda) = \mu_\varepsilon^{-1} \log \mathbb{E}_\varepsilon^{\text{gc}} \exp \{ \mu_\varepsilon \langle H, \mathcal{X}_\varepsilon \rangle + \lambda \cdot C_\varepsilon \}.$$

B2 proves on every compact real source chart and a fixed complex thickening the normally convergent decomposition

$$Q_\varepsilon(H, \lambda) = Q_\varepsilon^{(1)}(H, \lambda) + R_\varepsilon(H, \lambda),$$

where $Q_\varepsilon^{(1)}$ is the singleton/compound contribution and R_ε is the sum of connected hard-core trajectory clusters.

Assumption 1.1 (Admissible constraint chart). On a compact chart $\mathcal{K}$ the following hold.

- (i) the affine span of X is all of $\mathbb{R}^d$;
- (ii) finitely many product patches have a full-rank sum map and cover a positive fraction of the tilted one-particle activity;
- (iii) the B2 connected remainder is holomorphic on one complex neighbourhood of $\mathcal{K}$ and its first four derivatives satisfy the cluster bounds stated in B2.

Redundant constraints are removed before the chart is declared.

Lemma 1.2 (Typical-sector covariance for the full pressure). *There are $c, C > 0$ such that, uniformly on $\mathcal{K}$ and for all small ε ,*

$$cI_d \leq D_{\lambda\lambda}^2 Q_\varepsilon(H, \lambda) \leq CI_d.$$

The lower bound comes from the linearly many particles in the typical compound activity and is stable under the connected remainder.

Proof. The singleton Hessian equals the intensity-weighted second moment of X . The affine-span condition and compactness give a lower bound $2cI_d$. Differentiate the connected cluster series twice. Its terms with at least two labels are bounded by the absolutely summable B2 tree norm. On the chosen Boltzmann–Grad/source chart their operator norm is at most c after reducing the common chart once; no conditioning on a fixed low-particle sector is used. The upper bound follows from the same cluster majorant. $\square$

Corollary 1.3 (Exact finite source-dependent saddle). *For every target a in a compact subset of the mean image and every $H \in \mathcal{K}$, the equation*

$$D_\lambda Q_\varepsilon(H, \lambda_{H,\varepsilon}) = a$$

has a unique solution. The map $(H, a) \mapsto \lambda_{H,\varepsilon}$ is analytic in H , C^2 in a , and converges with its first two derivatives to the limiting saddle λ_H .

Proof. Apply the quantitative inverse-function theorem to Lemma 1.2. Normal convergence of the full B2 pressure and Cauchy estimates give convergence of the maps and their derivatives. $\square$

1.2. Regenerative smoothing with the connected remainder. Choose disjoint phase-space cells separated by more than the hard-core interaction range at time zero. In each cell expose a fixed number n_0 of marks. A cell is good when its marks lie in one of the full-rank patches from Assumption 1.1. Let M_ε be the number of good cells.

Lemma 1.4 (Conditional good-block minorization). *There are $c, p > 0$ and a sigma-field $\mathcal{G}_\varepsilon$ containing the outside configuration such that*

$$\mathbb{P}_{H,\lambda,\varepsilon}(M_\varepsilon < c\mu_\varepsilon) \leq e^{-c\mu_\varepsilon},$$

and, conditionally on $\mathcal{G}_\varepsilon$, every selected cell has a minorized full-rank mark-sum density

$$\mathcal{L}(S_j \mid \mathcal{G}_\varepsilon) \geq p g_{j,\varepsilon}(y) dy, \quad \sup_j \|g_{j,\varepsilon}\|_{W^{k,1}} \leq C_k.$$

The remainder of its conditional law is retained as an arbitrary probability measure.

Proof. Use cells whose enlarged collision cylinders are disjoint. The hard-core Gibbs specification and the B2 connected-source Radon–Nikodym derivative change the conditional activity in one such cell by factors bounded above and below uniformly on the chart. Coarea on the finite full-rank patches gives the density minorization. Reveal alternating cell families to recover conditional independence; a martingale Chernoff bound gives linearly many successful cells. Iterating over the finite colouring of the cell graph proves the exponential estimate. No complement of a local patch is declared smooth. $\square$

Theorem 1.5 (Full mixed lattice–continuous characteristic estimate). *Let t range in the fundamental domain of the lattice constraints and let u be the continuous Fourier variable. Uniformly on compact source/target charts,*

$$|\Phi_{H,\lambda,\varepsilon}(t, u)| \leq \begin{cases} \exp\{-c\mu_\varepsilon(|t|^2 + |u|^2)\}, & |t| + |u| \leq \delta, \\ e^{-c\mu_\varepsilon}, & \delta \leq |t| \leq \pi, \ |u| \leq R, \\ e^{-c\mu_\varepsilon} + C_k(1 + |u|)^{-k}, & |u| \geq R, \end{cases}$$

for every fixed k . The estimate is for the exact hard-sphere tilted law, not for an ideal compound proxy.

Proof. The central estimate is the Taylor expansion of the full complex pressure at the exact saddle, using Lemma 1.2. Lattice aperiodicity follows from the span-one particle-number activity and the finite mark lattice. On compact minor arcs, conditional good blocks have a Fourier modulus uniformly below one. At large continuous frequency, expand the minorized part in every good cell; choosing k cells and integrating their $W^{k,1}$ densities by parts gives the polynomial factor, while all remaining good cells give exponential

contraction. The event with too few good cells has probability $e^{-c\mu_\varepsilon}$ by Lemma 1.4. The conditional specifications already contain R_ε , so no later perturbative replacement is made. $\square$

1.3. The weighted shell integral. Write the exact constraint as $(L_\varepsilon, Y_\varepsilon)$, with lattice part L_ε and continuous extensive part Y_ε . Let

$$\mathcal{S}_\varepsilon = \{L_\varepsilon = \ell_\varepsilon, Y_\varepsilon - y_\varepsilon \in W_\varepsilon\},$$

where W_ε is a bounded Borel set whose diameter may be $o(\sqrt{\mu_\varepsilon})$ but need not be $o(1)$. Put $\Sigma_{H,\varepsilon} = D_{\lambda\lambda}^2 Q_\varepsilon(H, \lambda_{H,\varepsilon})$ and let $z = (Y - y)/\sqrt{\mu_\varepsilon}$.

Theorem 1.6 (Exact-saddle weighted local coefficient). *Uniformly on the admissible chart,*

$$\begin{aligned} & \mathbb{E}_\varepsilon^{\text{gc}} \left[e^{\mu_\varepsilon \langle H, \mathcal{X}_\varepsilon \rangle}; \mathcal{S}_\varepsilon \right] \\ &= \exp \{ \mu_\varepsilon Q_\varepsilon(H, \lambda_{H,\varepsilon}) - \lambda_{H,\varepsilon} \cdot (\ell_\varepsilon, y_\varepsilon) \} \frac{1 + o(1)}{(2\pi\mu_\varepsilon)^{r/2}} \int_{W_\varepsilon/\sqrt{\mu_\varepsilon}} e^{-\lambda_{H,\varepsilon}^c \cdot \sqrt{\mu_\varepsilon} z} \varphi_{\Sigma_{H,\varepsilon}^c}(z) dz. \end{aligned}$$

Here r is the number of lattice coordinates and Σ^c is the conditional continuous covariance after the lattice block is fixed. The factor involving λ^c may be replaced by one only when the original continuous shell has diameter $o(1)$.

Proof. Change measure to the exact $(H, \lambda_{H,\varepsilon})$ law. On the shell, the inverse likelihood is $\exp\{-\lambda^L \ell_\varepsilon - \lambda^c \cdot (y_\varepsilon + \sqrt{\mu_\varepsilon} z)\}$; retain its z dependence. Use Fourier inversion for the lattice coordinates and the density inversion for the continuous coordinates. Theorem 1.5 makes the minor and large arcs negligible relative to the Gaussian coefficient, while the central pressure expansion gives the displayed integral. Uniform covariance bounds supply domination for every declared shell scale. $\square$

Theorem 1.7 (Full microcanonical pressure and prepared LDP). *The conditioned log-Laplace functional satisfies*

$$Q^{\text{mc},a}(H) = \inf_\lambda \{ Q(H, \lambda) - \lambda \cdot a \} - \inf_\lambda \{ Q(0, \lambda) - \lambda \cdot a \},$$

with the saddle reoptimized at every H . The primitive shell laws satisfy the good prepared LDP obtained by conditioning the full B2 grand-canonical rate, and their Gaussian covariance is the corresponding Schur complement.

Proof. Apply Theorem 1.6 separately to the source numerator and the zero-source denominator. Their weighted Gaussian shell integrals have only subexponential logarithms; no unjustified pointwise cancellation is used. This gives the pressure formula. The upper bound follows from the full conditioned Laplace principle and exponential tightness. For the lower bound, use the B2 source-tilted recovery law at a regular point and the same weighted coefficient under that law, then approximate arbitrary finite-rate points by the B2 balance-preserving recovery theorem. Differentiating the exact saddle formula twice gives the Schur complement. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Every cross-paper input is typed in the round-nine dependency ledger. Repository proof and build gates are internal checks and do not replace independent external mathematical review.

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